ABASYN UNIVERSITY
Islamabad Campus

- **Name:** **Sajjal Tariq**
- **Roll no:** **10011**
- **Section:** **BSCS (sec A)**
- **Subject:** **Computer Organization Assembly Language**

Lab No 4 & 5

Task no 01:

Write a standalone assembly program that adds the following three words in AX register:

.DATA
array WORD 890Fh,1276h,0AF5Bh

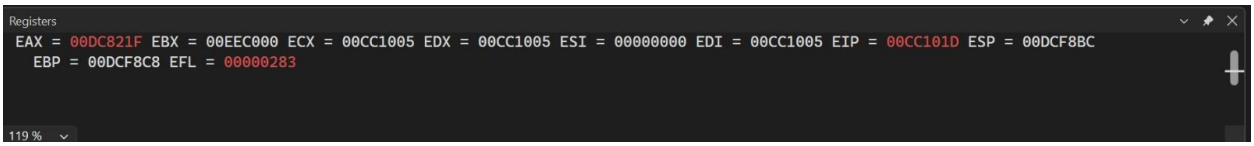
Does the sum fit in AX?

Program:

```
.386p
.model flat,stdcall
.data
array word 890Fh,1276h,0AF5Bh
.code
main proc
mov esi,0
add ax, array[esi*2]
inc esi
add ax, array[esi*2]
inc esi
add ax, array[esi*2]
main endp
end main
```

```
2    .386p
3    .model flat,stdcall
4    .data
5    array word 890Fh,1276h, 0AF5BH
6    .code
7    main proc
8    mov esi, 0
9    add ax,array[esi*2]
10   inc esi
11   add ax, array[esi*2]
12   inc esi
13   add ax, array[esi*2]
14   main endp
15   end main
```

Output:



Task no 02:

Modify the program 1 to accumulate the sum in EAX register. [Hint: Use of movzx/movsx]

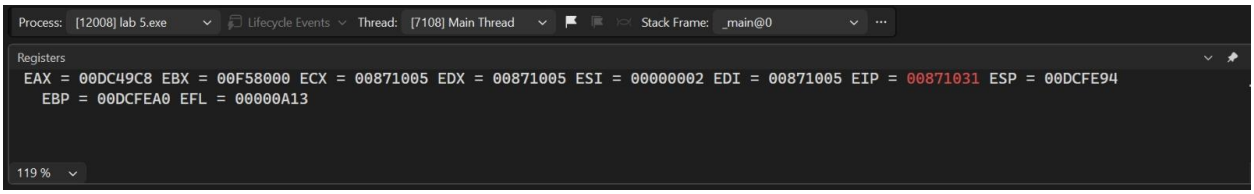
Program:

```
.386p
.MODEL flat, stdcall
.data
array word 890fh,1276h,0af5bh
main proc
mov esi, 0
add ax, array[esi*2]
inc esi
add ax, array[esi*2]
inc esi
add ax,array[esi*2]
mov ax , eax
main endp
```

end main

```
1   .386p
2   .MODEL flat, stdcall
3   .data
4   array word 890fh,1276h,0af5bh
5   main proc
6
7   mov esi, 0
8
9   add ax, array[esi*2]
10
11  inc esi
12
13  add ax, array[esi*2]
14
15  inc esi
16
17  add ax,array[esi*2]
18  mov ax , eax
19  main endp
20  end main
```

Output:



Task no 03:

Modify the program 2 by employing;

Indirect addressing - base

Indirect addressing - indexed

Program:

.386p
.MODEL flat, stdcall
.data
array word 890fh,1276h,0af5bh
main proc
mov esi, 0
add ax, array[esi*2]
inc esi

add ax, array[esi*2]
inc esi
add ax,array[esi*2]
mov ax , eax
mov eax, OOFSET array
mov eax, [ax*2]
main endp
end main

```
2       .386p
3       .MODEL flat, stdcall
4       .data
5       array word 890fh,1276h,0af5bh
6       main proc
7
8       mov esi, 0
9
10      add ax, array[esi*2]
11
12      inc esi
13
14      add ax, array[esi*2]
15
16      inc esi
17
18      add ax,array[esi*2]
19      mov ax , eax
20      mov eax, OOFSET array
21      mov eax, [ax*2]
22      main endp
23      end main
```

Output:

